

PGA™

Prediction Governance Architecture — Prediction Governance Layer

Official Prediction Governance Architecture for Human Forecasting, AI Systems,
Predictive Models, Prediction Markets, Autonomous Agents, Organizations,
Decision Systems, Collective Intelligence, and Future Predictive Environments

PGA™ serves as the official OOF® prediction governance architecture used for Prediction Object & Purpose governance, Prediction Methodology governance, Prediction Basis governance, Prediction Assumption governance, Prediction Formation, Probability & Uncertainty governance, Prediction Feasibility governance, Collective Prediction governance where applicable, Prediction Communication & Reliance governance, Prediction Monitoring & Revision, Prediction Outcome & Performance governance, and Prediction Record & Continuity governance across human forecasting, AI systems, predictive models, prediction markets, autonomous agents, organizations, decision systems, collective intelligence environments, and future predictive systems.

PGA™ does not make predictions. PGA™ defines the governance architecture required to ensure that predictions become explicitly defined, methodologically documented, basis-grounded, assumption-visible, properly formed, uncertainty-bounded, feasibility-aware, appropriately composed where collective prediction applies, responsibly communicated and relied upon, monitored and revisable, legitimately associated with governed outcomes, performance-assessable, and historically reconstructable throughout their complete lifecycle.

PGA™ consists of 12 Core Parent Standards and 60 Core Modules forming the primary Prediction Governance backbone within the OOF® ecosystem.

Additional compatible standards, bridge standards, and specialized Prediction Governance standards may extend the architecture when new Prediction Governance spaces are identified. However, the standards listed below form the official core Prediction Governance structure of PGA™.

Canonical Definition

PGA™ — Prediction Governance Architecture is the governance architecture that defines how a Prediction Object is established, predictive purpose and boundaries are defined, methodology is documented, Prediction Basis and assumptions are governed, a specific prediction is formed, probability and uncertainty are represented, feasibility is assessed, multiple predictions are collectively composed where applicable, predictions

are communicated and relied upon, changing conditions trigger review or revision, predictions are compared with governed outcomes, and complete historical record remains identifiable and reconstructable throughout the Prediction Lifecycle.

Core Question

How can predictions remain methodologically defined, traceable, uncertainty-bounded, feasibility-aware, appropriately relied upon, revisable, performance-assessable, and historically reconstructable throughout their complete lifecycle?



Governance Flow

Prediction Object & Purpose → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → Probability & Uncertainty → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

Core Parent Standards

1. Prediction Object & Purpose Standard (POPS)

Governed Space: Prediction Object & Predictive Purpose

Core Question: What exactly is being predicted, for what purpose, over what horizon, and within what predictive boundary?

2. Prediction Methodology Standard (PMES)

Governed Space: Prediction Methodology

Core Question: By what documented and reproducible methodology is the prediction legitimately constructed?

3. Prediction Basis Standard (PBAS)

Governed Space: Prediction Basis

Core Question: What information, data, observations, evidence references, and signals legitimately form the basis of the prediction?

4. Prediction Assumption Standard (PAS)

Governed Space: Prediction Assumptions

Core Question: What assumptions must hold for the prediction and its stated probability, confidence, feasibility, or expected outcome to remain meaningful?

5. Prediction Formation Standard (PFOS)

Governed Space: Prediction Formation

Core Question: How are methodology, Prediction Basis, assumptions, and applicable conditions transformed into a specific governed prediction?

6. Prediction Probability & Uncertainty Standard (PPUS)

Governed Space: Predictive Probability & Uncertainty

Core Question: How should probability, uncertainty, confidence, ranges, and conditionality be represented without overstating what the prediction actually establishes?

7. Prediction Feasibility Standard (PFES)

Governed Space: Prediction Feasibility

Core Question: Is the predicted outcome realistically achievable under the stated assumptions, constraints, time horizon, Prediction Basis, and relevant conditions?

8. Collective Prediction Governance Standard (CPGS)

Governed Space: Collective Prediction

Core Question: Under what conditions may predictions originating from multiple participants, models, agents, markets, or forecasting mechanisms be combined and represented as a legitimate collective prediction?

9. Prediction Communication & Reliance Standard (PCRS)

Governed Space: Prediction Communication & Reliance

Core Question: How may a prediction be communicated, interpreted, and relied upon without exceeding what its methodology, uncertainty, feasibility, intended purpose, and governed conditions legitimately support?

10. Prediction Monitoring & Revision Standard (PMRS)

Governed Space: Prediction Monitoring & Revision

Core Question: When changing information or conditions materially affect a prediction, when must it be preserved, reviewed, revised, superseded, or withdrawn?

11. Prediction Outcome & Performance Standard (POFPS)

Governed Space: Prediction Outcome & Predictive Performance

Core Question: Once a sufficiently governed outcome becomes available, how should the prediction be associated with that outcome and its predictive performance legitimately determined over time?

12. Prediction Record & Continuity Standard (PRCS)

Governed Space: Prediction Record & Historical Continuity

Core Question: How must a prediction and its complete history remain identifiable, reconstructable, versioned, and historically continuous across formation, revision, supersession, outcome, performance assessment, retirement, and lifecycle re-entry?

Compatible OOF® Architectures

PGA™ interoperates with multiple OOF® architectures governing complementary governance spaces.

GOA™ — Governance Architecture provides the constitutional governance layer defining how governance itself is structured, bounded, delegated, and maintained.

ORA™ — Operational Reality Architecture governs what is actually occurring in Operational Reality, while PGA™ governs predictions concerning possible, expected, probable, conditional, or anticipated future outcomes.

VALIDOS™ — Validation Governance Architecture governs whether a validation basis sufficiently supports a Validation Determination, while PGA™ governs the Prediction Lifecycle and the methodological conditions under which predictions are formed, represented, used, revised, and assessed.

INTEGROS® — Integrity Governance Architecture governs integrity across objects, states, evidence, and processes, while PGA™ governs prediction-specific methodology, basis, assumptions, formation, uncertainty, feasibility, lifecycle, performance, and historical continuity.

ADIT® governs evidential observation, preservation, verification, examination, and auditability, while PGA™ governs which information forms the Prediction Basis and how the Prediction Lifecycle remains governable.

AGA™ — Accountability Governance Architecture governs authority, responsibility, accountability, attribution, and decision relationships surrounding predictive activity and reliance.

AIG® — AI Governance Architecture governs behavioral governance of AI systems, while PGA™ governs predictions produced, communicated, interpreted, or used by AI systems where those predictions enter a governed Prediction Lifecycle.

ASGA™ — Autonomous Systems Governance Architecture governs autonomous-system operation and action, while PGA™ governs predictions used or produced within autonomous predictive and decision environments.

CLIA® governs cognition-related governance spaces, while PGA™ governs the prediction as a distinct governed object regardless of whether its formation depends on human, machine, collective, or hybrid cognition.

SIMULOS® — Simulation Governance Architecture governs simulation purpose, structure, execution, outputs, and lifecycle, while PGA™ governs predictions that may use simulation outputs as part of their methodology or Prediction Basis without treating simulation and prediction as the same governed object.

PGA™ serves as the common Prediction Governance architecture supporting the broader OOF® ecosystem by ensuring that predictions become governed objects rather than unstructured claims about future outcomes.

Architectural Position

PGA™ governs the complete Prediction Lifecycle from definition of the Prediction Object to preservation of the complete historical Prediction Record.

The architecture progresses through:

Prediction Object & Purpose → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → Probability & Uncertainty → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

Together these standards govern what is being predicted and why, how the prediction is methodologically constructed, what information and assumptions support it, how the prediction is formed, how probability and uncertainty are represented, whether the predicted outcome is feasible, how multiple predictions may be collectively composed, how predictions are communicated and relied upon, how they respond to changing conditions, how they are compared with governed outcomes, and how their complete historical lifecycle remains reconstructable.

Collective Prediction is conditional rather than mandatory.

The single-predictor path may therefore proceed:

Prediction Feasibility → Communication & Reliance

while collective environments may proceed:

Prediction Feasibility → Collective Prediction → Communication & Reliance

Architecture Position ≠ Mandatory Processing Through Every Optional Governance Space.

PGA™ remains distinct from the systems that actually generate predictive calculations.

Prediction Model / Human Forecaster / Agent / Market / Predictive Mechanism → Governed Prediction Lifecycle under PGA™

PGA™ governs predictions. It does not make predictions.

Core Governed Prediction Spaces

PGA™ establishes twelve distinct governable Prediction spaces:

Prediction Object & Predictive Purpose → what exactly is being predicted, why, over what horizon, and within what boundary.

Prediction Methodology → by what documented and reproducible methodology the prediction is constructed.

Prediction Basis → which information, data, observations, evidence references, and signals form the basis of the prediction.

Prediction Assumptions → which assumptions must remain visible and sufficiently governed for the prediction to remain meaningful.

Prediction Formation → how methodology, Prediction Basis, assumptions, and applicable conditions become a specific Governed Prediction.

Predictive Probability & Uncertainty → how probability, uncertainty, confidence, ranges, and conditionality are represented.

Prediction Feasibility → whether the predicted outcome is realistically achievable under relevant conditions and constraints.

Collective Prediction → how multiple predictions may legitimately become a governed collective prediction where collective composition applies.

Prediction Communication & Reliance → how a prediction may be represented, interpreted, communicated, and relied upon.

Prediction Monitoring & Revision → how changing information and conditions trigger preservation, review, revision, supersession, or withdrawal.

Prediction Outcome & Predictive Performance → how a prediction becomes associated with a sufficiently governed outcome and how Predictive Performance is legitimately assessed.

Prediction Record & Historical Continuity → how the prediction and its complete lifecycle history remain identifiable, versioned, reconstructable, and continuous through time.

Each space answers a different methodological question.

Each can fail independently.

Their separation is therefore foundational to the PGA™ architecture.

Cross-Architecture Boundary Lock

PGA™ is intentionally bounded to prevent duplication with adjacent OOF® architectures and prediction-generating systems.

PGA™ ↔ ORA™

ORA™: What actually happened in Operational Reality?

PGA™: What was predicted, under what governed predictive conditions, and how does that prediction relate to a sufficiently governed outcome?

Prediction Outcome Association ≠ Outcome Determination.

Prediction Governance ≠ Operational Reality Governance.

PGA™ ↔ VALIDOS™

PGA™: Has the Prediction Lifecycle been sufficiently governed for the applicable predictive purpose?

VALIDOS™: Does the total validation basis sufficiently support a Validation Determination and intended reliance?

Prediction Governance ≠ Validation Determination.

PGA™ ↔ INTEGROS®

PGA™: governs Prediction Object & Purpose, methodology, basis, assumptions, formation, probability and uncertainty, feasibility, collective prediction, communication and reliance, monitoring and revision, outcome and performance, and record continuity.

INTEGROS®: governs integrity across objects, states, evidence, and processes.

Prediction Governance ≠ General Integrity Governance.

PGA™ ↔ ADIT®

PGA™: governs which information forms the Prediction Basis and how that basis participates in the Prediction Lifecycle.

ADIT®: governs evidential observation, preservation, verification, examination, and auditability.

Prediction Basis ≠ Evidence Governance.

PGA™ ↔ AGA™

PGA™: governs the methodological Prediction Lifecycle.

AGA™: governs who holds authority, responsibility, accountability, and attribution for prediction-related decisions, actions, reliance, and failures.

Prediction Methodology ≠ Accountability Allocation.

PGA™ ↔ AIG®

PGA™: governs the prediction as a governed object.

AIG®: governs AI behavioral integrity and related behavioral governance.

AI-Generated Prediction ≠ AI Behavioral Governance.

PGA™ ↔ ASGA™

PGA™: governs predictions used or produced within predictive systems.

ASGA™: governs autonomous-system authority, action, constraints, intervention, escalation, and operational governability.

Prediction Governance ≠ Autonomous Action Governance.

PGA™ ↔ SIMULOS®

PGA™: governs predictions.

SIMULOS®: governs simulations.

A simulation may contribute to Prediction Methodology or Prediction Basis, but the resulting prediction remains a distinct governed object.

Simulation Output ≠ Governed Prediction by Default.

PGA™ ↔ Prediction Models and Forecasting Mechanisms

PGA™: governs the Prediction Lifecycle.

Prediction Model / Forecasting Mechanism: produces or contributes to predictive calculation, estimation, inference, judgment, market signal, or forecast generation.

Prediction Governance ≠ Prediction Generation.

PGA™ governs predictions. It does not make predictions.

Architectural Origin

PGA™ evolved from the foundational methodological problem that predictions increasingly influence decisions without a common governance layer defining what the prediction actually is, how it was constructed, what information and assumptions support it, what its probability means, whether its predicted outcome is feasible, how it may be relied upon, how it should change when conditions change, and how its historical performance should be interpreted.

Traditional predictive environments often move directly from:

Data / Information → Model / Forecaster → Prediction → Decision

without explicitly governing the Prediction Lifecycle itself.

As AI systems, prediction markets, autonomous agents, collective intelligence systems, probabilistic models, automated decision systems, and large-scale forecasting environments expand, prediction

generation becomes increasingly separated from prediction governance.

PGA™ therefore establishes a dedicated governance architecture around the prediction as an independent methodological object.

Its foundational principles are distributed primarily across:

Prediction Object & Purpose → Prediction Methodology → Prediction Basis → Prediction Assumptions → Prediction Formation → Probability & Uncertainty → Prediction Feasibility → Collective Prediction where applicable → Communication & Reliance → Monitoring & Revision → Outcome & Performance → Prediction Record & Continuity

The architecture establishes the Governed Prediction independently of the mechanism that generated it and preserves its methodological meaning throughout the complete Prediction Lifecycle.

Architectural Structure

Architecture: PGA™ — Prediction Governance Architecture

Category: Governance & Enforcement

Subcategory: Prediction Governance

Core Parent Standards: 12 Core Modules: 60 Architecture Model: Lifecycle-Based · Modular · Methodology-Neutral · Technology-Neutral Primary Governance Domain: Prediction Governance

The architecture deliberately separates governable spaces that can independently fail.

A prediction may:

have an undefined Prediction Object or purpose; use an undocumented methodology; rely upon an insufficient or misleading Prediction Basis; contain hidden or invalid assumptions; be improperly formed; misrepresent probability, uncertainty, confidence, range, or conditionality;

describe an outcome that is possible but operationally infeasible; present a collective signal without legitimate collective composition; be communicated more strongly than its methodology supports; be relied upon beyond its legitimate purpose or conditions; remain unchanged after material conditions change; be evaluated against an inappropriate or insufficiently governed outcome; appear accurate because of information advantage, outcome proximity, intervention, or selective evaluation rather than demonstrated predictive capability; lose its historical versions, revisions, withdrawals, outcomes, performance states, or lifecycle continuity.

PGA™ treats these not as one generic prediction quality problem , but as distinct Prediction Governance spaces requiring distinct methodological controls.

Recursive Lifecycle Principle

PGA™ does not end when a prediction is first formed.

A Governed Prediction may continue to change throughout its lifecycle.

During that period:

New information may enter the Prediction Basis.

Assumptions may change.

Probability and uncertainty may change.

Feasibility conditions may change.

The prediction may be communicated and relied upon.

Reliance may influence Operational Reality.

Review may become necessary.

The prediction may be revised.

A later prediction state may supersede an earlier state.

The prediction may be withdrawn.

A sufficiently governed outcome may become available.

Predictive Performance may become assessable or reassessable.

The Prediction Record may be retired or archived.

A historical Prediction Record may later re-enter governed use.

PGA™ therefore closes recursively through PMRS, POFPS, and PRCS:

Governed Prediction → Monitoring → Review → Revision / Supersession / Withdrawal → Outcome Association → Predictive Performance → Historical Prediction Record → Continuity

Where material change occurs:

Historical Prediction State → New Governed Prediction Version → Preserved Historical Continuity

Where a historical Prediction Record later becomes relevant:

Historical / Inactive Prediction Record → Re-Entry Evaluation → Re-Entry Classification → Re-Entered Prediction Record State or New Prediction Formation

This makes prediction change, evaluation, and historical continuity part of Prediction Governance rather than uncontrolled forecasting-data maintenance.

Foundational Architecture Principles

PGA™ governs predictions. It does not make predictions.

A prediction without a documented methodology is an opinion, not a governed prediction.

Prediction Basis ≠ Evidence Governance.

A hidden assumption cannot remain a governed assumption.

Probability ≠ Confidence.

Probability ≠ Certainty.

Confidence ≠ Certainty.

High Probability ≠ High Confidence.

Same Probability Value ≠ Same Predictive Meaning.

Probability Representation Requires Methodological Context.

Possible ≠ Plausible ≠ Feasible ≠ Probable.

Feasible ≠ Probable.

Multiple Predictors ≠ Independent Predictions.

Number of Participants ≠ Information Diversity.

A collective signal is not automatically a governed prediction.

Market-Implied Probability ≠ Governed Probability Estimate.

Market-Implied Probability ≠ Population Belief.

A prediction must not become stronger in communication than it is in methodology.

Communicable ≠ Reliance-Eligible.

Valid Initial Probability ≠ Valid Continuing Probability.

Revision ≠ Historical Replacement.

Supersession ≠ Deletion.

Prediction Outcome Association ≠ Outcome Determination.

Observed Accuracy ≠ Demonstrated Predictive Skill.

Predictive Success ≠ Predictive Capability.

Information Advantage ≠ Predictive Capability.

Outcome Knowledge ≠ Predictive Capability.

Outcome Knowledge ≠ Outcome Control.

Decision Proximity ≠ Decision Authority.

Predictive Accuracy ≠ Trading Profitability.

Trading Profitability ≠ Predictive Reliability.

Current Prediction State ≠ Complete Prediction History.

The current prediction state must not erase the historical prediction record from which it evolved.

A governed prediction may be wrong. It must not be methodologically undefined.

Canonical Principle

PGA™ does not govern the future itself, generate predictions, establish Operational Reality, determine evidence validity, allocate accountability, govern general AI behavior, control autonomous action, or treat simulation, market price, model output, collective signal, or probability value as a Governed Prediction by default. PGA™ governs the methodological conditions under which a Prediction Object and predictive purpose are defined, Prediction Methodology is documented, Prediction Basis and assumptions are governed, a specific prediction is formed, probability and uncertainty are represented, feasibility is assessed, collective prediction is governed where applicable, communication and reliance remain bounded, changing conditions trigger review and revision, predictions are associated with sufficiently governed outcomes, Predictive Performance is assessed, and the complete Prediction Record remains historically identifiable, versioned, reconstructable, and continuous throughout its lifecycle.

PGA™ governs the prediction — not the future.

PGA™ Parent Standards

[Prediction Object & Purpose Standard \(POPS\)](#)

[Prediction Methodology Standard \(PMES\)](#)

[Prediction Basis Standard \(PBAS\)](#)

[Prediction Assumption Standard \(PAS\)](#)

[Prediction Formation Standard \(PFOS\)](#)

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[Prediction Outcome & Performance Standard \(POFPS\)](#)

[Prediction Record & Continuity Standard \(PRCS\)](#)

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